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Polynomials over the integers

Level 5 Advanced · Algebra

What integer coefficients force: divisibility of differences and periodic points, the binomial basis, Gauss, Eisenstein and root-size irreducibility, roots of unity, and Schur's theorem on prime divisors.

By the end you can: You will be able to use a - b — P(a) - P(b), difference tables and the binomial basis to decide which values an integer polynomial can take; You will be able to prove irreducibility with Gauss's lemma, Eisenstein (after a shift) and estimates on the size of roots; You will be able to use cyclotomic polynomials and Schur's theorem, combined with the Chinese remainder theorem, in olympiad proofs.

Lesson

Is there a polynomial P with integer coefficients such that $P(2) = 5$ and $P(9) = 17$? No. For every $k \geq 1$ the number $9^k - 2^k$ is divisible by $9 - 2 = 7$, so $P(9) - P(2)$ is a multiple of 7, while $17 - 5 = 12$ is not. This one-line argument is the first of several facts in this unit that hold for polynomials with integer coefficients and fail for polynomials with rational or real coefficients. We assume L3-AL1 (Vieta, the factor theorem) and the modular arithmetic of L4-NT1 throughout. Write $\mathbb{Z}[x]$ for the set of polynomials with integer coefficients.

Differences of values

Theorem 1. If $P \in \mathbb{Z}[x]$ and a, b are integers, then $a - b \mid P(a) - P(b)$. In particular, $P(n) \bmod m$ depends only on $n \bmod m$.
Proof. $P(a) - P(b) = \sum_k c_k(a^k - b^k)$ and $a^k - b^k = (a - b)(a^{k-1} + a^{k-2}b + \cdots + b^{k-1})$.

Two further facts about $\mathbb{Z}[x]$ are used constantly. If $D \in \mathbb{Z}[x]$ is *monic*, long division of $P \in \mathbb{Z}[x]$ by D never divides by anything except 1, so the quotient and remainder lie in $\mathbb{Z}[x]$. Hence if $P \in \mathbb{Z}[x]$ vanishes at the integers $r_1, \ldots, r_k$, then $P(x) = (x - r_1) \cdots (x - r_k)Q(x)$ with $Q \in \mathbb{Z}[x]$.

Example 1. A polynomial P with integer coefficients satisfies $P(P(P(5))) = 5$. Prove that $P(5) = 5$.

Solution.
We know nothing about P except one composite value, so the only usable fact is Theorem 1, applied along the orbit. Put $x_0 = 5$ and $x_{i+1} = P(x_i)$, so $x_3 = x_0$, and let $d_i = x_{i+1} - x_i$. Note that $x_4 = P(x_3) = P(x_0) = x_1$, so $d_3 = d_0$.
By Theorem 1, $d_i = x_{i+1} - x_i$ divides $P(x_{i+1}) - P(x_i) = d_{i+1}$. Hence $d_0 \mid d_1 \mid d_2 \mid d_3 = d_0$, and so $|d_0| = |d_1| = |d_2| = d$, say. But $d_0 + d_1 + d_2 = x_3 - x_0 = 0$, and a sum of three numbers each equal to $\pm d$ is an odd multiple of d. Hence $d = 0$, that is, $P(5) = x_1 = x_0 = 5$.

The same argument with k steps proves the general statement, which is worth remembering.

Theorem 2 (periodic points). Let $P \in \mathbb{Z}[x]$, $x_0 \in \mathbb{Z}$ and $x_{i+1} = P(x_i)$. If $x_k = x_0$ for some $k \geq 1$, then $x_2 = x_0$.
Proof. With $d_i = x_{i+1} - x_i$ as above, the sequence (d_i) has period k and $d_i \mid d_{i+1}$. If some $d_i = 0$ then all later d_j are 0, so by periodicity all $d_i = 0$ and $x_1 = x_0$. Otherwise $|d_0| \leq |d_1| \leq \cdots \leq |d_k| = |d_0|$, so all $|d_i|$ are equal. Since $d_0 + \cdots + d_{k-1} = 0$, the signs are not all equal, so $d_{i+1} = -d_i$ for some i, which means $x_{i+2} = x_i$. Then $x_{j+2} = x_j$ for all $j \geq i$, and

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by periodicity for all j, in particular $j = 0$.

Integer-valued polynomials and the binomial basis

Theorem 1 is necessary for integer coefficients but not sufficient, and the right tool for deciding exactly which values a polynomial in $\mathbb{Z}[x]$ can take is the basis $\binom{x}{k} = \frac{x(x-1)\cdots(x-k+1)}{k!}$. Write $\Delta P(x) = P(x+1) - P(x)$. Pascal's rule gives $\Delta\binom{x}{k} = \binom{x}{k-1}$.

Theorem 3. Every polynomial P of degree at most n (with rational, real or complex coefficients) can be written uniquely as $P(x) = \sum_{k=0}^{n} c_k \binom{x}{k}$, and $c_k = \Delta^k P(0)$. Moreover:

1. $P(m) \in \mathbb{Z}$ for every integer m if and only if every c_k is an integer, if and only if $P(0), P(1), \ldots, P(n)$ are integers;
2. $P \in \mathbb{Z}[x]$ if and only if $k! \mid c_k$ for every k.

Proof. Since $\binom{x}{k}$ has degree k, these polynomials form a basis, and applying Δ^j and setting $x = 0$ kills every term except c_j. For (1): if $P(0), \ldots, P(n)$ are integers, so are all differences $\Delta^j P(0)$; conversely $\binom{m}{k}$ is an integer for every integer m (for $m < 0$ note $\binom{m}{k} = (-1)^k\binom{k-m-1}{k}$). For (2): if $k! \mid c_k$ then $c_k\binom{x}{k} = \frac{c_k}{k!}x(x-1)\cdots(x-k+1) \in \mathbb{Z}[x]$. Conversely, if $P \in \mathbb{Z}[x]$ has degree n and leading coefficient a, comparing leading coefficients gives $c_n = n!\,a$, and $P - a\,x(x-1)\cdots(x-n+1)$ lies in $\mathbb{Z}[x]$ and has smaller degree; induct.

Note that for a polynomial in $\mathbb{Z}[x]$ of any degree, $c_k = \Delta^k P(0)$ for $k \le n$ depends only on $P(0), \ldots, P(n)$. Hence values $v_0, \ldots, v_n$ are taken at $0, \ldots, n$ by some $P \in \mathbb{Z}[x]$ if and only if $k! \mid \Delta^k v(0)$ for $k = 0, \ldots, n$.

Example 2. Find all integers t for which some polynomial with integer coefficients satisfies $P(0) = 3$, $P(1) = 1$, $P(2) = 3$, $P(3) = 15$ and $P(4) = t$.

Solution.
First try Theorem 1 on each pair: $4 \mid t - 3$, $3 \mid t - 1$, $2 \mid t - 3$ and the rest hold automatically, so $t \equiv 7 \pmod{12}$. This is necessary, but we do not yet know it is sufficient. The difference table is:

$$\begin{array}{ccccccccc}
3 & & 1 & & 3 & & 15 & & t \\
& -2 & & 2 & & 12 & & t-15 & \\
& & 4 & & 10 & & t-27 & & \\
& & & 6 & & t-37 & & & \\
& & & & t-43 & & & &
\end{array}$$

So $c_0 = 3$, $c_1 = -2$, $c_2 = 4$, $c_3 = 6$, $c_4 = t - 43$. The conditions $1 \mid c_1$, $2 \mid c_2$, $6 \mid c_3$ hold, and $24 \mid c_4$ is needed. Hence $t \equiv 43 \equiv 19 \pmod{24}$. Conversely, for such t the polynomial $3 - 2\binom{x}{1} + 4\binom{x}{2} + 6\binom{x}{3} + (t-43)\binom{x}{4}$ lies in $\mathbb{Z}[x]$ by Theorem 3 and takes the required values. The answer is $t \equiv 19 \pmod{24}$; in particular $t = 7$ passes every test from Theorem 1 and still fails.

Irreducibility

A polynomial in $\mathbb{Z}[x]$ is *primitive* if the highest common factor of its coefficients is 1.

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Theorem 4 (Gauss's lemma). The product of two primitive polynomials is primitive. Hence if $f \in \mathbb{Z}[x]$ factors as a product of two nonconstant polynomials in $\mathbb{Q}[x]$, it factors as a product of two nonconstant polynomials in $\mathbb{Z}[x]$ of the same degrees; if f is monic, both factors can be taken monic.
Proof. Let $g = \sum b_i x^i$, $h = \sum c_j x^j$ be primitive and p a prime. Take the least i with $p \nmid b_i$ and the least j with $p \nmid c_j$. The coefficient of x^{i+j} in gh is $b_i c_j$ plus terms each containing some $b_{i'}$ with $i' < i$ or $c_{j'}$ with $j' < j$, so it is not divisible by p. For the second part, write a rational factorisation as $f = \frac{r}{s} g_0 h_0$ with g_0, h_0 primitive; then $sf = r g_0 h_0$, and comparing contents gives $\frac{r}{s} \in \mathbb{Z}$.

Theorem 5 (Eisenstein). Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ and p a prime with $p \mid a_0, \ldots, a_{n-1}$, $p \nmid a_n$ and $p^2 \nmid a_0$. Then f is irreducible over $\mathbb{Q}$.
Proof. By Theorem 4 suppose $f = gh$ with $g = \sum_{i \le r} b_i x^i$, $h = \sum_{j \le s} c_j x^j$ in $\mathbb{Z}[x]$ and $r, s \ge 1$. Exactly one of b_0, c_0 is divisible by p, say $p \mid b_0$. Since $p \nmid a_n = b_r c_s$, $p \nmid b_r$, so there is a least $i \le r < n$ with $p \nmid b_i$. Then $a_i = b_i c_0 + b_{i-1} c_1 + \cdots \equiv b_i c_0 \not\equiv 0 \pmod p$, contradicting $p \mid a_i$.

Eisenstein often applies only after the substitution $x \mapsto x+1$ (or $x \mapsto x+a$), which does not change irreducibility. When no prime works, look at the sizes of the complex roots instead.

Theorem 6 (small constant term). Let $f \in \mathbb{Z}[x]$ be monic with $|f(0)|$ prime, and suppose every complex root z of f has $|z| > 1$. Then f is irreducible over $\mathbb{Q}$.
Proof. If $f = gh$ with g, h monic, nonconstant and in $\mathbb{Z}[x]$, then $|g(0)|$ is the product of the moduli of the roots of g, so $|g(0)| > 1$, and similarly $|h(0)| > 1$. These are integers, so $|f(0)| = |g(0)||h(0)|$ is a product of two integers greater than 1, which is not prime.

Example 3. Determine all integers $n \ge 2$ for which $x^n + x + 2$ is irreducible over $\mathbb{Q}$.
Solution.
Eisenstein at 2 fails because the coefficient of x is 1. The constant term is the prime 2, so Theorem 6 is the thing to try: we need to know whether a root can satisfy $|z| \le 1$. If $|z| \le 1$ then $|z^n + z| \le |z|^n + |z| \le 2$, and $z^n + z = -2$ forces equality throughout: $|z| = 1$ and z^n, z point in the same direction, so $z^n = z$ and $2z = -2$, that is $z = -1$ and $(-1)^n = -1$.
Hence if n is even, every root has $|z| > 1$ and the polynomial is irreducible by Theorem 6. If n is odd, $(-1)^n - 1 + 2 = 0$, so $x + 1$ is a factor and, as $n \ge 2$, the polynomial is reducible. The answer is: exactly the even n. Note that the equality case of the triangle inequality is where the parity of n enters; a solution that only says $|z^n + z| \le 2$ proves nothing.

Roots of unity

The cyclotomic polynomials are defined by $x^n - 1 = \prod_{d \mid n} \Phi_d(x)$, so Φ_n is the monic polynomial whose roots are the primitive nth roots of unity. By induction on n and division by monic polynomials, $\Phi_n \in \mathbb{Z}[x]$. For a prime p, $\Phi_p(x) = 1 + x + \cdots + x^{p-1}$, and

$$\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{p-1} x^{p-2} + \cdots + \binom{p}{1}$$

satisfies Eisenstein at p, so Φ_p is irreducible. (Every Φ_n is irreducible over $\mathbb{Q}$; this is a theorem of Gauss whose proof for composite n is longer, and I will not use it.)

Theorem 7. Let p be prime, $\zeta = e^{2\pi i/p}$, and $a_0, \ldots, a_{p-1}$ rationals with $a_0 + a_1\zeta + \cdots + a_{p-1}\zeta^{p-1} = 0$. Then $a_0 = a_1 = \cdots = a_{p-1}$.

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Proof. Let $A(x) = \sum a_i x^i$ and $D = \gcd(A, \Phi_p)$ in $\mathbb{Q}[x]$. Since $D = UA + V\Phi_p$ for some $U, V \in \mathbb{Q}[x]$, $D(\zeta) = 0$, so D is a nonconstant factor of the irreducible Φ_p. Hence $\Phi_p \mid A$, and as $\deg A \le p-1$, $A = c\,\Phi_p$.

We also need the fundamental theorem of symmetric polynomials: a symmetric polynomial with integer coefficients in $z_1, \dots, z_n$ is an integer polynomial in the elementary symmetric functions. So if $z_1, \dots, z_n$ are the roots of a monic $P \in \mathbb{Z}[x]$, every such expression is an integer.

Prime divisors of values

Theorem 8 (Schur). If $P \in \mathbb{Z}[x]$ is nonconstant, infinitely many primes divide at least one of the nonzero numbers among $P(1), P(2), P(3), \dots$
Proof. If $P(0) = 0$, every prime p divides $P(p)$, and $P(p) \neq 0$ for all but finitely many p. Otherwise let $c = P(0) \neq 0$ and suppose only the primes $p_1, \dots, p_r$ occur; put $N = p_1 \cdots p_r$. Every nonconstant term of $P(|c|Nx)$ is divisible by cNx, so $P(|c|Nx) = c\,(1 + Nx\,Q(x))$ with $Q \in \mathbb{Z}[x]$, and $1 + NxQ(x)$ is nonconstant. It equals ± 1 for only finitely many x, so choose a positive integer x with $|1 + NxQ(x)| > 1$ and a prime q dividing it. Then $P(|c|Nx) \neq 0$ and $q \mid P(|c|Nx)$, but $1 + NxQ(x) \equiv 1 \pmod{p_i}$ for every i, so q is a new prime. (We use $|c|$ rather than c so that the argument $|c|Nx$ is a positive integer.)

Example 4. Let $P \in \mathbb{Z}[x]$ be nonconstant and k a positive integer. Prove that there is a positive integer n such that $P(n)$ has at least k distinct prime factors.

Solution.
Schur gives many primes, but each comes with its own argument a; we need one n for all of them at once, so we combine the congruences with the Chinese remainder theorem. By Theorem 8 choose distinct primes $p_1, \dots, p_k$ and positive integers $a_1, \dots, a_k$ with $p_i \mid P(a_i)$. By CRT there is n_0 with $n_0 \equiv a_i \pmod{p_i}$ for all i, and then every $n = n_0 + m p_1 \cdots p_k$ satisfies the same congruences. By Theorem 1, $P(n) \equiv P(a_i) \equiv 0 \pmod{p_i}$ for each i. Finally P has finitely many roots, so we may choose m with $n > 0$ and $P(n) \neq 0$, and then $P(n)$ has the k distinct prime factors $p_1, \dots, p_k$.
The condition $P(n) \neq 0$ is the step most scripts omit: without it, "$p_i \mid P(n)$ for all i" says nothing.

Common traps.

- Treating $a - b \mid P(a) - P(b)$ as a complete test for which values are possible. It is not (Example 2); use differences and Theorem 3.
- Dividing by a non-monic polynomial and assuming the quotient has integer coefficients.
- Applying Eisenstein without checking the two end conditions: p must not divide the leading coefficient, and p^2 must not divide the constant term.
- Proving that f has no rational root and concluding that f is irreducible. This is valid only in degrees 2 and 3.
- In root-size arguments, using $\le$ where the lemma needs $<$, and not analysing the equality case.

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How to spot this in a contest

The phrase "polynomial with integer coefficients" is a signal to use Theorem 1 at once, on every pair of arguments in sight. My order of attack is:

1. Values at several integers, or iterates such as $P(P(a))$: differences of arguments divide differences of values; for orbits, Theorem 2.
2. Which values are possible, or "$P(n)$ is an integer for all n": the binomial basis and difference tables.
3. Irreducibility: Eisenstein, then Eisenstein after $x \mapsto x + a$, then root sizes against a prime constant term; factor as gh with g, h monic in $\mathbb{Z}[x]$ and compare $g(0)$, $h(0)$, or $g(a)$, $h(a)$ at a cleverly chosen a.
4. Sums of powers of a root of unity, or counting modulo a prime: Theorem 7 and the factorisation of $x^p - 1$.
5. "Infinitely many primes" or "$P(n)$ divides something for all n": Schur, then CRT to combine primes.

Problems at BMO2 and IMO level usually need two of these together, for example Schur followed by Theorem 1 in the form $P(n + p) \equiv P(n) \pmod{p}$.

Practice

Try each problem before turning to the hints and solutions at the back.

1. ⋆⋆ Let P be a polynomial with integer coefficients and $a \neq b$ integers such that $P(a)P(b) = -(a - b)^2$. Prove that $P(a) + P(b) = 0$.

2. ⋆⋆ How many polynomials P with integer coefficients and degree at most 4 satisfy $P(0), P(1), P(2), P(3), P(4)$ $\{0, 1, 2, 3\}$?

3. ⋆⋆⋆ Let p be an odd prime. Prove that

$$f(x) = x^{p-2} + 2x^{p-3} + 3x^{p-4} + \cdots + (p-2)x + (p-1)$$

is irreducible over $\mathbb{Q}$.

4. ⋆⋆⋆ Find all polynomials P with integer coefficients such that $P(n)$ divides n^n for every positive integer n.

5. ⋆⋆⋆⋆ Let P be a monic polynomial with integer coefficients, all of whose complex roots have absolute value 1. Prove that every root of P is a root of unity.

6. ⋆⋆⋆⋆ Let P be a nonconstant polynomial with integer coefficients and k a positive integer. Prove that there are k consecutive positive integers $n + 1, n + 2, \ldots, n + k$ such that each of $P(n + 1), \ldots, P(n + k)$ is divisible by the square of some prime.

Contest ladder

Real past problems that use this idea, easiest first. Open them on the official paper or the AoPS wiki, or online at kevinvariant08.github.io.

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1. USAMO 1974 Problem 1 ⋆⋆
2. IMO 1993 Problem 1 ⋆⋆⋆
3. USAMO 1976 Problem 5 ⋆⋆⋆
4. BMO2 2013/14 Problem 3 ⋆⋆⋆
5. BMO2 2018/19 Problem 3 ⋆⋆⋆
6. BMO2 2011/12 Problem 4 ⋆⋆⋆
7. USAMO 2012 Problem 4 ⋆⋆⋆⋆
8. USAMO 1995 Problem 4 ⋆⋆⋆⋆
9. IMO 2006 Problem 5 ⋆⋆⋆⋆

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Hints

1. Compare the power of each prime p in $P(a)$, $P(b)$ and $a-b$, using $a-b \mid P(a)-P(b)$.
2. Each choice of the five values gives exactly one polynomial of degree at most 4. Use Theorem 1 to cut down the choices, then test the fourth difference.
3. Compute $(x-1)^2 f(x)$, then substitute $x \mapsto x+1$.
4. Write $P(x) = x^k Q(x)$ with $Q(0) \neq 0$, and look at $Q(q)$ for a large prime q.
5. For each $k \geq 1$ consider the monic polynomial whose roots are the kth powers of the roots of P. How many such polynomials can there be?
6. Take an irreducible factor R of P; then R and R' have no common factor. Find primes p and integers b with $p^2 \mid R(b)$, then use CRT.

Solutions

Problem 1

Let $u = P(a)$, $v = P(b)$ and $d = a - b \neq 0$. Then $uv = -d^2 \neq 0$, so $u, v \neq 0$, and by Theorem 1, $d \mid u - v$.

Fix a prime p and let α, β, δ be the exponents of p in u, v, d. From $uv = -d^2$ we get $\alpha + \beta = 2\delta$. If $\alpha \neq \beta$, then the exponent of p in $u - v$ is $\min(\alpha, \beta)$, which is less than $\frac{\alpha+\beta}{2} = \delta$; this contradicts $d \mid u - v$. Hence $\alpha = \beta = \delta$.

This holds for every prime, so $|u| = |v| = |d|$. Since $uv = -d^2 < 0$, u and v have opposite signs, and therefore $u = -v$, that is, $P(a) + P(b) = 0$.

Problem 2

Answer: 6

By Lagrange interpolation each choice of values $v_i = P(i)$ gives exactly one polynomial of degree at most 4, so we count the tuples $(v_0, \ldots, v_4)$ that come from a polynomial in $\mathbb{Z}[x]$. By Theorem 3 (and the remark after it) this happens if and only if $k! \mid \Delta^k v(0)$ for $k \leq 4$.

First we use Theorem 1. Since $4 \mid v_4 - v_0$, $v_4 = v_0 = a$, say. Since $3 \mid v_3 - v_0$ and $3 \mid v_4 - v_1$, both v_1 and v_3 are congruent to a modulo 3; since $2 \mid v_3 - v_1$, they have the same parity. If $a \in \{1, 2\}$ this forces $v_1 = v_3 = a$; if $a \in \{0, 3\}$ it forces $v_1 = v_3 = b$ with $b \in \{0, 3\}$. Finally $v_2 \equiv a \pmod 2$. This leaves twelve tuples (a, b, c, b, a), with $b = a$ when $a \in \{1, 2\}$.

The fourth difference is $\Delta^4 v(0) = v_4 - 4v_3 + 6v_2 - 4v_1 + v_0 = 2a - 8b + 6c$, which must be divisible by 24. For $a = b = 1$, $c \in \{1, 3\}$ gives 0 or 12; for $a = b = 2$, $c \in \{0, 2\}$ gives -12 or 0; for $a = 0$, $(b, c) \in \{0, 3\} \times \{0, 2\}$ gives $0, 12, -24, -12$; for $a = 3$, $(b, c) \in \{0, 3\} \times \{1, 3\}$ gives $12, 24, -12, 0$. So exactly six tuples survive: the four constant tuples, $(0, 3, 0, 3, 0)$ and $(3, 0, 3, 0, 3)$.

For each of these the lower differences also satisfy the conditions: constants are clear, and $(0, 3, 0, 3, 0)$ has differences $c_0, \ldots, c_4 = 0, 3, -6, 12, -24$, divisible by $0!, 1!, 2!, 3!, 4!$ respectively, so $3\binom{x}{1} - 6\binom{x}{2} + 12\binom{x}{3} - 24\binom{x}{4} \in \mathbb{Z}[x]$; the last tuple comes from 3 minus this polynomial. The answer is 6. (Theorem 1 alone would have allowed 12.)

Problem 3

Write $f(x) = \sum_{k=1}^{p-1} k\, x^{p-1-k}$. Then

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$$(x-1)f(x) = \sum_{k=1}^{p-1} k\,x^{p-k} - \sum_{k=1}^{p-1} k\,x^{p-1-k} = x^{p-1} + x^{p-2} + \cdots + x - (p-1),$$

since the coefficient of x^j for $1 \le j \le p-1$ is $(p-j) - (p-1-j) = 1$ and the constant term is $-(p-1)$. Multiplying by $x-1$ again,

$$(x-1)^2 f(x) = (x^p - x) - (p-1)(x-1) = x^p - px + (p-1).$$

Substituting $x+1$ for x:

$$x^2 f(x+1) = (x+1)^p - p(x+1) + p - 1 = \sum_{k=2}^{p} \binom{p}{k} x^k,$$

because the terms $1 + px$ from the binomial expansion cancel with $-px - p + p - 1$. Hence

$$g(x) = f(x+1) = x^{p-2} + \binom{p}{p-1} x^{p-3} + \cdots + \binom{p}{3} x + \binom{p}{2}.$$

For $2 \le k \le p-1$, p divides $\binom{p}{k} = \frac{p!}{k!(p-k)!}$, since p divides the numerator and not the denominator. The leading coefficient is 1, and the constant term $\binom{p}{2} = p \cdot \frac{p-1}{2}$ is not divisible by p^2 because $\frac{p-1}{2}$ is an integer less than p. By Eisenstein, g is irreducible over $\mathbb{Q}$. If $f = uv$ with u, v nonconstant, then $g(x) = u(x+1)v(x+1)$ would be a factorisation of g, so f is irreducible too. (For $p = 3$, $f = x + 2$ and the statement is trivial; the argument covers this case.)

Problem 4

The answer is $P(x) = \pm 1$, $\pm x$ and $\pm x^2$. These work: $n^k \mid n^n$ for $k \le 2$ when $n \ge 2$, and for $n = 1$ we need $P(1) \mid 1$, which holds since $P(1) = \pm 1$ for all six polynomials.

Conversely, suppose P works. Since $n^n \neq 0$, $P(n) \neq 0$ for all $n \ge 1$, so P is not the zero polynomial; write $P(x) = x^k Q(x)$ with $Q \in \mathbb{Z}[x]$ and $c = Q(0) \neq 0$. Then $Q(n) \mid P(n) \mid n^n$ for all $n \ge 1$.

Let q be a prime with $q > |c|$. Every prime factor of $Q(q)$ divides q^q, so $Q(q) = \pm q^j$ for some $j \ge 0$. By Theorem 1, $Q(q) \equiv Q(0) = c \pmod q$, and $q \nmid c$ because $0 < |c| < q$. Hence $j = 0$ and $Q(q) = \pm 1$. This holds for infinitely many primes q, so the polynomial $Q(x)^2 - 1$ has infinitely many roots and is zero; hence Q is the constant 1 or -1.

So $P(x) = \pm x^k$, and the condition at $n = 2$ is $2^k \mid 4$, so $k \le 2$. Together with the check above, the solutions are exactly ± 1, $\pm x$, $\pm x^2$.

Problem 5

Let $n = \deg P$ and let $z_1, \ldots, z_n$ be the roots of P, with multiplicity, so $P(x) = \prod (x - z_i)$. For each positive integer k define $P_k(x) = \prod_{i=1}^{n} (x - z_i^k)$.

The coefficient of x^{n-j} in P_k is $(-1)^j e_j(z_1^k, \ldots, z_n^k)$, where e_j is the jth elementary symmetric function. As a function of $z_1, \ldots, z_n$ this is a symmetric polynomial with integer coefficients, so by the fundamental theorem of symmetric polynomials it is an integer polynomial in $e_1(z), \ldots, e_n(z)$, which are (up to sign) the coefficients of P and hence integers. So $P_k \in \mathbb{Z}[x]$ and is monic of degree n.

Since $|z_i^k| = 1$, $e_j(z_1^k, \ldots, z_n^k)$ is a sum of $\binom{n}{j}$ numbers of absolute value 1, so its absolute value is at most $\binom{n}{j}$. Hence each coefficient of P_k is an integer in a fixed bounded range, and there are only finitely many possible polynomials P_k as k varies.

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Each P_k has at most n roots, and z_1^k is a root of P_k. So all the numbers $z_1, z_1^2, z_1^3, \dots$ lie in the union of the root sets of finitely many polynomials, which is a finite set. Hence $z_1^k = z_1^l$ for some $k < l$, and since $z_1 \neq 0$, $z_1^{l-k} = 1$. The same argument applies to every root.

Problem 6

Let $R \in \mathbb{Z}[x]$ be a nonconstant factor of P that is irreducible over $\mathbb{Q}$ (by Theorem 4 we may take it in $\mathbb{Z}[x]$). Its derivative R' is nonzero of smaller degree, so a common factor of R and R' in $\mathbb{Q}[x]$ would be a factor of R of degree between 1 and $\deg R - 1$, which does not exist. Hence $\gcd(R, R') = 1$ in $\mathbb{Q}[x]$, and there are $U, V \in \mathbb{Q}[x]$ with $UR + VR' = 1$. Clearing denominators, $U_1 R + V_1 R' = M$ for some $U_1, V_1 \in \mathbb{Z}[x]$ and a positive integer M.

By Schur's theorem infinitely many primes divide some value $R(a)$ with a a positive integer. Choose k distinct such primes $p_1, \dots, p_k$, none dividing M, and positive integers a_i with $p_i \mid R(a_i)$. Then $p_i \nmid R'(a_i)$, since otherwise $p_i \mid U_1(a_i)R(a_i) + V_1(a_i)R'(a_i) = M$.

For integers a and h we have $R(a+h) = R(a) + hR'(a) + h^2 S(a,h)$ with S an integer polynomial, because $(a+h)^m = a^m + ma^{m-1}h + h^2(\dots)$. Since $p_i \nmid R'(a_i)$, choose an integer t_i with $\frac{R(a_i)}{p_i} + t_i R'(a_i) \equiv 0 \pmod{p_i}$ and put $b_i = a_i + t_i p_i$. Then $R(b_i) \equiv R(a_i) + t_i p_i R'(a_i) \equiv 0 \pmod{p_i^2}$, so $p_i^2 \mid R(b_i) \mid P(b_i)$.

The moduli $p_1^2, \dots, p_k^2$ are pairwise coprime, so by CRT there is a positive integer n with $n + i \equiv b_i \pmod{p_i^2}$ for $i = 1, \dots, k$. By Theorem 1, $P(n+i) \equiv P(b_i) \equiv 0 \pmod{p_i^2}$, so each $P(n+i)$ is divisible by the square of the prime p_i.

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